

FROM CAMERA TO DECISION

How Your Driver Monitoring System Works

A complete walkthrough with real numbers

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Step-by-step, one frame at a time

How This Works

Your Driver Monitoring System processes every video frame the same way. One camera frame comes in. It goes through six stages. At the end, you know if the driver is ACTIVE, DROWSY, or in DANGER. This document walks through those six stages with a real example — actual numbers, actual calculations — so you see exactly what happens.

The example driver has already been watched for 10 seconds, so the system knows what their 'normal' looks like. Now we're processing one frame where their eyes are drooping. Let's follow that frame all the way to the decision.

Why six stages? Because no single measurement is perfect. MediaPipe finds the eyes. EAR measures them geometrically. The CNNs read them as pictures. The ensemble checks if anything is weird. PERCLOS adds time. The state machine decides. Together, they're bulletproof.

Stage 1: MediaPipe Finds the Face

What happens

The camera captures one frame. MediaPipe Face Mesh runs on this frame and places 468 reference points — landmarks — all over the face in 3D space. These aren't random; they're anatomically consistent. Landmark 33 is always the corner of the right eye (your right when looking at the person). Landmark 70 is always the corner of the left eye. And so on.

Why this works

You can't measure an eye if you don't know where it is. MediaPipe solves that in milliseconds on a CPU. It's fast because Google designed it for phones. It's accurate because it's trained on hundreds of thousands of faces. And it's consistent — the landmarks sit in the same spots on every face, every time.

Our example frame — MediaPipe's output

Our driver's face is visible in this frame. MediaPipe drops 468 landmarks. The ones we care about are the six around each eye:

Right eye: landmarks 33, 160, 158, 133, 153, 144 Left eye: landmarks 362, 385, 387, 263, 373, 380

We extract their pixel coordinates:

RIGHT EYE:	LEFT EYE: P1 = (33, 40)	P1 = (70, 40)
[inner corner] P2 = (39, 35)	P2 = (76, 34)	[top-left]
P3 = (51, 35)	P3 = (88, 34)	[top-right]
P4 = (94, 40)	[outer corner] P5 = (51, 43)	P5 = (88, 44)
[bottom-right] P6 = (39, 43)	P6 = (76, 44)	[bottom-left]

These six points form a shape around each eye. In this frame, both eyes are starting to droop — they're closing. Let's measure how much.

Stage 2: Calculate EAR — The Eye Aspect Ratio

What it measures

EAR is one number that tells you how open an eye is. It compares the height of the eye to its width. Open eye → tall, narrow shape → high EAR. Closing eye → the height shrinks → EAR falls.

The formula (both eyes separately)

For the right eye, using the six landmarks:

$$\text{EAR}_R = \frac{(|P2 - P6| + |P3 - P5|)}{2 \times |P1 - P4|} = \frac{\text{vertical1} + \text{vertical2}}{2 \times \text{horizontal width}}$$

Plug in the numbers:

$$\text{Vertical pair 1: } |35 - 43| = 8 \quad \text{Vertical pair 2: } |35 - 43| = 8 \quad \text{Horizontal width: } |33 - 57| = 24 \quad \text{EAR}_R = (8 + 8) / (2 \times 24) = 16 / 48 = 0.167$$

For the left eye:

$$\text{Vertical pair 1: } |34 - 44| = 10 \quad \text{Vertical pair 2: } |34 - 44| = 10 \quad \text{Horizontal width: } |70 - 94| = 24 \quad \text{EAR}_L = (10 + 10) / (2 \times 24) = 20 / 48 = 0.417$$

What the numbers mean

Right eye: 0.167. That's low — the right eye is drooping.

Left eye: 0.417. That's high — the left eye is still mostly open.

Average them: $\text{EAR}_{\text{avg}} = (0.167 + 0.417) / 2 = 0.292$.

Recall: this driver's baseline (when alert) was $\text{EAR} = 0.30$. This frame at 0.292 is just barely below normal. But one eye is clearly drooping. The CNNs will look closer.

Key insight

EAR is dirt cheap to compute — just a few subtractions and one division. That's why it runs on every frame. But it varies per person and can be fooled by glasses or head angle. That's why the CNNs are there.

Stage 3: CNN Cascade — Visual Confirmation

What happens

The system crops the eye region from the image. Two neural networks classify it:

CNN V1 (tiny, fast): runs on every frame. Takes a 32×32 grayscale crop. Outputs a probability the eye is open.

CNN V2 (powerful, slow): runs only if V1 is unsure. Takes a 96×96 color crop. More accurate.

Our example — CNN cascade

V1 gets the right-eye crop. It says: $P(\text{open}) = 0.42$. That's ambiguous — between 0.3 and 0.7. So the cascade triggers V2.

V2 gets the same right-eye crop but at higher resolution. It says: $P(\text{open}) = 0.19$. It's confident the eye is closed.

The verdict for this frame's right eye: CLOSED. The left eye cascade says OPEN.

Why two networks?

V1 is fast and good at blur. V2 is powerful and good at occlusion. One is cheap, one is expensive. Running both only when needed gives you speed on normal frames and accuracy on hard frames.

Stage 4: The Ensemble Votes — Is This Abnormal?

The question

EAR and the CNNs told us the eye state. But they don't tell us whether this is normal for THIS driver. That's the ensemble's job. It asks: 'Is $\text{EAR} = 0.292$ unusual for someone whose baseline is 0.30?'

The baseline (calibration)

Remember: during the first 10 seconds, the system watched this driver and learned:

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Mean EAR = 0.30 Standard deviation = 0.03
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This is this driver's personal 'normal.'

Four models vote

Each model has its own way of thinking about 'normal.' Here's what they vote on 0.292:

Model 1: K-Means (weight 1)

This model found two clusters in the baseline: open-centroid at 0.30, closed-centroid at 0.12. Distance from 0.292 to each:

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Distance to open: |0.292 - 0.30| = 0.008 ← closer Distance to closed: |
0.292 - 0.12| = 0.172
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0.292 is very close to the open cluster. K-Means votes: NORMAL. → Vote: 0 (no anomaly).

Model 2: GMM — Gaussian Mixture (weight 1)

How many standard deviations away from the open mean?

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z-score = (0.292 - 0.30) / 0.03 = -0.008 / 0.03 = -0.27
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Only 0.27σ below normal — well within the open-eye bell curve. Probability this belongs to normal: ~79%. GMM votes: NORMAL. → Vote: 0.

Model 3: Isolation Forest (weight 2)

How easily does a random tree isolate this point? 0.292 sits in the dense normal region, so many random splits are needed to isolate it. It's NOT an anomaly. → Vote: 0.

Model 4: One-Class SVM (weight 2)

Does 0.292 fall inside the learned boundary around normal? The boundary is roughly $\text{EAR} > 0.24$. And $0.292 > 0.24$. Yes, inside. → Vote: 0.

The tally

K-Means:	0 votes	GMM:	0 votes	IsoForest:	0 votes	(weight
2) OC-SVM:	0 votes	(weight 2)	TOTAL:			0
out of 6						

All four models agree: this single frame is not abnormal. The ensemble score is 0/6. This is important: one frame close to baseline doesn't trigger anything. Only sustained closure does, which is the job of PERCLOS.

Stage 5: PERCLOS — Accumulating Time

What it measures

PERCLOS is the percentage of time the eyes are closed over the last 30 seconds. At 30 FPS that's 900 frames. One frame closed means nothing. But if many frames over the last 30 seconds were closed, PERCLOS rises.

How it builds

Every frame, the system records: closed or open (from EAR and CNN). These feed a rolling window. $\text{PERCLOS} = (\text{closed frames in window} / \text{total frames}) \times 100\%$.

Our example — the window

This is frame 523 in the video. The rolling window holds frames 24–923 (the last 900 frames, i.e., the last 30 seconds). Over that window, the driver's right eye has been drooping repeatedly, accumulating closed frames. Left eye mostly open.

The count:

Closed frames in window: 396	Total frames:	900	PERCLOS =
$(396 / 900) \times 100 = 44.0\%$			

What it means

44% is in the DROWSY band (20–57%). The driver has been keeping their eyes closed for a big chunk of the last 30 seconds. Not sleeping — but drowsy.

PERCLOS ignores blinks. A normal blink is <100ms, < 1 frame at 30 FPS. Over 30 seconds, normal blinks add almost nothing to the percentage. Only sustained closures move PERCLOS meaningfully.

Stage 6: The State Machine Decides

The rule

PERCLOS sets the band. The ensemble score confirms it. Here's the decision logic:

if PERCLOS < 20%:	state = ACTIVE	vehicle = full speed	if 20% ≤
PERCLOS < 57%:	state = DROWSY	vehicle = 50% speed	if PERCLOS ≥ 57%:
state = DANGER	vehicle = emergency stop (after 10 continuous seconds)		

Our example — the decision

PERCLOS = 44% → falls in the DROWSY band (20–57%).

Ensemble score = 0/6 → no strong anomaly signal on this instant, but...

The fact that PERCLOS climbed to 44% means many frames **over time** were closed. That's what matters. One instantaneous anomaly doesn't trip the system, but accumulated closure does.

Decision: DROWSY.

Vehicle response: velocity reduced to 50%. If the driver can't shake this off after a few seconds, velocity drops more. If PERCLOS climbs above 57% and stays there for 10 continuous seconds, the emergency stop triggers autonomously.

The Complete Picture — One Frame

Here's everything that just happened, in order, in one frame:

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1. MediaPipe detects 468 landmarks in the camera frame.    ↓ 2. EAR is
computed from 6 landmarks per eye:    Right: 0.167 (drooping)    Left:
0.417 (open)    Average: 0.292    ↓ 3. CNN cascade classifies each eye:
Right eye: CNN V1 unsure (0.42) → V2 fires → CLOSED (0.19)    Left eye:
OPEN    ↓ 4. Ensemble votes on EAR_avg = 0.292 vs baseline 0.30:    K-
Means: normal (0 votes)    GMM: normal (0 votes)    IsoForest: normal (0
votes)    OC-SVM: normal (0 votes)    Total: 0/6 (no anomaly *this
instant*)    ↓ 5. PERCLOS accumulates the closed-frame count over 30
seconds:    44.0% of the last 900 frames were closed    ↓ 6. State machine
decides:    PERCLOS 44% → DROWSY band (20–57%)    Ensemble 0/6 → not
alarming *this instant*    But accumulated closure over time is the key.
DECISION: DROWSY    RESPONSE: vehicle speed reduced to 50%
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All of this happens in roughly 50 milliseconds. The system doesn't stop you — it warns and restricts. The driver has time to shake it off. If they don't, after 10 seconds of continuous DANGER (PERCLOS > 57%), the car does an autonomous emergency stop.

Why it's robust

No single measurement decides. EAR can be fooled by sunglasses. The CNN can be fooled by extreme lighting. The ensemble instantly would flag false alarms. PERCLOS is temporal — it sums over 30 seconds, so one blink or glance never triggers anything. The state machine is the referee.

The system is designed so every component has a job and every flaw is covered by another. That's how you get from noisy sensors and imperfect models to a decision you can trust.